

Write-Up

What Didn't Work, Trade-offs, and What I'd Build Next

What Didn't Work

Client discovery for private companies was the hardest problem. Company websites almost never list customers publicly. The pipeline runs dedicated searches like “[company] supplies to which OEM” which works for well-covered companies like Brakes India, but would find almost nothing for a smaller private company without press coverage. No clean fix exists without paid databases like Bloomberg or Dun & Bradstreet.

Financial data for private companies has the same dependency on coverage. Brakes India worked because it has a CRISIL rating and Tofler data. A smaller company with no credit rating would produce a nearly empty financial section. MCA filings are the right long-term answer but parsing them programmatically is non-trivial.

Citation consistency is not fully clean. The financial highlights section has Claude computing figures like “36.8% revenue growth” internally from Yahoo Finance numbers rather than citing them from an explicit source. The system should either source pre-computed figures or omit them entirely.

Search variability. Tavily returns different results on different runs for the same query, meaning two runs on the same company can produce slightly different outputs. A production system needs deterministic retrieval.

Trade-offs

Cost vs coverage		8 Tavily searches per company costs ~\$0.10 and significantly improves client and product coverage vs a single generic query. Scaling to hundreds of companies daily would need caching or a cheaper retrieval layer.
Latency depth	vs	Each run takes 25–40 seconds, mostly Tavily calls running sequentially. Parallelising with <code>asyncio</code> would cut this under 10 seconds. Fine for batch overnight runs, too slow for an interactive product.
Accuracy completeness	vs	Temperature 0.0 with strict grounding produces accurate, cited output but leaves gaps when retrieval misses something. Higher temperature fills gaps but risks invented clients or wrong figures. For PE analysts, an empty field is better than a wrong one.

What I'd Build Next

Parallel search calls — the biggest quick win, cutting latency from 30 seconds to under 10 with no quality loss.

MCA filing integration — for Indian private companies, MCA filings contain audited financials. A dedicated parser would solve the private company financial gap properly.

Smarter client retrieval — target supply chain databases, tender documents, and OEM supplier portals instead of general web search. Much higher signal for client relationships.

PDF rendering — convert the cited Markdown into a clean one-pager PDF with citations in footnotes, making it directly usable by an analyst.

Evaluation framework — define ground truth companies with known facts and automatically score each run on coverage and precision, enabling systematic iteration instead of manual eyeballing.